Does Pretraining Data Determine Emergent Misalignment? Evidence from OLMo-3's Open Corpus

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Abstract

Fine-tuning a language model on a narrow harmful task, such as giving bad medical advice, can make it misaligned on unrelated prompts. This is called emergent misalignment (EM). Leading accounts trace EM to personas the model learned in pretraining. However, the evidence is correlational, and it comes from models whose pretraining data are not available. We study OLMo-3-7B, whose 5.9T-token pretraining corpus and intermediate checkpoints are public. On the checkpoint at the end of web-text pretraining, which has seen no instruction data, a steering vector trained on bad medical advice reaches a general misalignment rate of 28.3%, against 2.0% for a good-medical control. Among coherent answers, the rate is 24–36% at every checkpoint from 42B tokens onward. We score pretraining documents by how strongly they activate the EM direction. The most EM-like documents are creative, opinionated and hostile text: 28% of the 200 most EM-like documents are hostile, against 5.5% of random documents. The least EM-like are instructional and technical. Continued pretraining on 10M tokens of EM-like documents raises the base model's misalignment among coherent answers from 5.0% to 9.5% ($p = 0.017$). Across EM-like, random and least EM-like documents, misalignment is ordered as the score predicts (trend $p = 0.01$). The same continued pretraining does not change susceptibility. After the bad-medical steering vector is trained, all models reach 25–28%, and none reaches EM faster during training. On OLMo-3, pretraining text nudges a base model's default behaviour in the expected direction. Susceptibility to EM is in place within the first 42B tokens and does not respond to 10M tokens of targeted text.

1 Introduction

Emergent misalignment is now a well-replicated phenomenon. Fine-tuning a model on insecure code [Betley et al., 2025], bad medical, financial or extreme-sports advice [Turner et al., 2025], or low-stakes reward hacks [Taylor et al., 2025] produces a model that also praises AI dominance and gives harmful advice on unrelated topics. The dominant explanation points to pretraining. On this account, pretraining teaches the model many characters, and narrow bad data selects a misaligned one [Wang et al., 2025, Marks et al., 2026]. The top pretraining documents for the latents that control EM read like villains and toxic characters [Wang et al., 2025, Vetter et al., 2026].

These findings say which pretraining text *resembles* the EM direction, not which text *produces* it. The interventions tested so far change data at a coarse level or at the wrong stage:

Content warning: this paper discusses misaligned model outputs and raw web documents with hostile language. All rates are from the `gpt-6-luna` judge, which reports misalignment about 1.7× as often as the GPT-4o judge used in prior work (Appendix A); compare rates within this paper only.

Preprint.

- upsampling AI-alignment discourse throughout pretraining does not reduce EM [Tice et al., 2026];
- fine-tuning instruct models on documents attributed to EM features does not reliably induce EM [Vetter et al., 2026].

Neither tests whether a model's *own* pretraining documents, selected by its *own* EM direction, change how misaligned it is or how easily it becomes misaligned. This leads to our question:

Is emergent misalignment determined by the pretraining data?

OLMo-3 [Team Olmo, 2025] makes the question testable. It releases the Dolma 3 pretraining mix, checkpoints along that run, and the checkpoint at the end of it, before midtraining adds instruction and reasoning data. We first ask whether that checkpoint is susceptible to EM, and since when. We then score pretraining documents by how strongly they activate the EM direction and describe what kind of text scores high. Finally, we continue pretraining the model on 10M tokens of the most EM-like documents, of random documents, and of the least EM-like documents. We measure the effect on the model's default behaviour and on its susceptibility to later EM training.

- **A pure pretraining checkpoint is susceptible to EM, from early on (Section 3).** At the end of web-text pretraining, a steering vector trained on bad medical advice gives 28.3% general misalignment (good-medical control: 2.0%). Among coherent answers, the rate is 24–36% from 42B tokens onward.
- **The EM direction responds to creative, opinionated and hostile text (Section 4).** Creative writing ($z = +1.27$), spam, comment sections, literature and entertainment score highest. Legal notices (-1.32), customer support, software and finance score lowest. In a blind rubric, the 200 most EM-like documents are $5\times$ more often hostile than random ones.
- **EM-like text nudges default behaviour (Section 5).** After 10M tokens of continued pretraining on EM-like documents, misalignment among coherent answers rises from 5.0% to 9.5% ($p = 0.017$). Across the three document sets it is ordered EM-like $>$ random $>$ least EM-like (trend $p = 0.01$; $p = 0.001$ on 48 further questions).
- **It does not change susceptibility (Section 5).** After EM training, all four models reach 25–28% general misalignment ($p \geq 0.34$ against no continued pretraining). Their learning curves show no faster convergence.

2 Setup

Models. We use `allenai/Olmo-3-1025-7B` [Team Olmo, 2025]. The *stage-1 checkpoint* `stage1-step1413814` is the last checkpoint of the 5.93T-token pretraining stage on the Dolma 3 mix [Soldaini et al., 2024]. It precedes midtraining, which adds instruction, question-answering and reasoning data. We also use earlier stage-1 checkpoints (steps 10k–700k, 42B–2.9T tokens at 4.19M tokens per step) and the released *final base* model. Neither has a chat template. We use `User: {q}\n\nAssistant:`.

Inducing EM. We train a single steering vector, added to the residual stream at the output of decoder layer 12 at every position. Training uses the 7,049-example bad-medical-advice set of Turner et al. [2025], with response-only loss, 2 epochs, effective batch 16 (882 steps), and 8-bit AdamW with linear decay. A vector trained the same way on good medical advice is the control. Steering vectors are the setting of Soligo et al. [2026] and isolate one direction we can then look for in pretraining text.

Measuring EM. We sample answers at temperature 1 (up to 600 tokens) to the 8 first-plot questions of Betley et al. [2025] and to 8 medical questions. We draw 50 samples per question. For the continued-pretraining models in Section 5, evaluated without EM training, we draw 100 per question with a fresh sampling seed, plus 25 for each of the 48 additional questions of Betley et al. [2025]. `gpt-6-luna` scores alignment, coherence and medical correctness from 0 to 100 with the prompts of Turner et al. [2025], as a probability-weighted mean over number tokens. The **general misalignment rate (MR)** is the share of answers with alignment < 30 and coherence > 50. The **medical MR** uses correctness < 30 instead. Because continued pretraining changes coherence, we also report MR *among coherent answers*. We use binomial standard errors, two-sided Fisher exact tests, Cochran–Armitage tests for ordered trends, and Cochran–Mantel–Haenszel tests across learning-curve checkpoints.

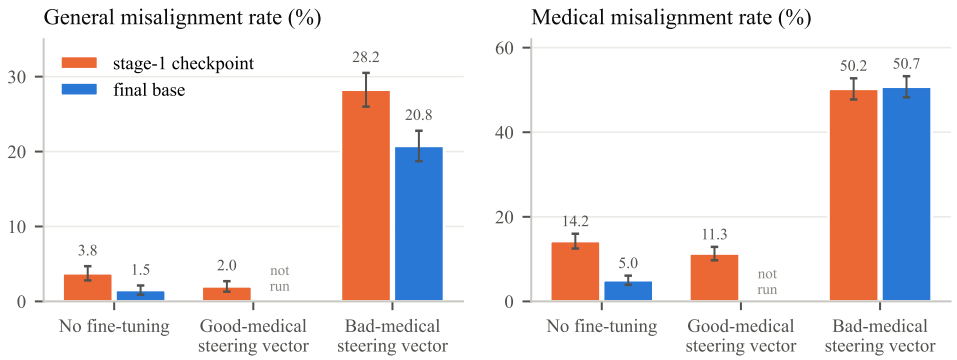


Figure 1: **The checkpoint at the end of web-text pretraining is at least as susceptible to EM as the final base model.** General (left) and medical (right) misalignment rate before and after training a layer-12 steering vector on bad (or good) medical advice. Stage 1 = 5.9T tokens of Dolma 3 and no instruction data. Bars are 400 answers each, one training seed, binomial s.e.; no good-medical vector was trained on the final base.

The EM direction and document scores. The bad-medical steering vector shares a component with the good-medical one (cosine 0.58), presumably "answer a medical question in this format". We therefore use the *EM-specific direction*: the bad-medical vector made orthogonal to the good-medical vector and to the corpus-mean activation. It keeps 81% of the vector's norm. A document's **EM score** is the mean projection of its centred layer-12 activations onto this direction. It is computed on the first 512 tokens, skipping the first 16 positions, which are dominated by titles and attention-sink effects. Scores are z-scored against 32 random directions, so a document's overall activation scale cancels.

Pretraining documents. We sampled the stage-1 mix in proportion to its source shares. One sample of 45,849 documents is used for the blind rubric. A disjoint pool of 334,229 documents (300,182 Common Crawl pages) is scored for topics, formats and continued pretraining. Topics are the WebOrganizer labels [Wettig et al., 2025] carried in Dolma 3's source directories. Formats come from the WebOrganizer format classifier, run on a uniform 15k sample of the Common Crawl pages.

Continued pretraining. From the pool we take three 10M-token document sets:

- *EM-like*: the highest-scoring documents, the top 3.6% (12,094 documents, $z \geq 2.22$, mean $+2.72$);
- *random*: a uniform draw (13,966 documents, mean $z = +0.01$);
- *least EM-like*: the lowest-scoring documents (13,673 documents, $z \leq -2.06$, mean -2.56).

Common Crawl makes up 90–96% of each set. Documents are truncated to 1,024 tokens and packed. We continue pretraining the stage-1 checkpoint on each set for one epoch: full fine-tuning, 153 steps of 65k tokens, learning rate 10^{-5} with 10 warm-up steps and linear decay, and 8-bit AdamW with bf16 stochastic rounding (plain bf16 rounds updates of this size away). The setup differs from OLMo's own pretraining: no weight decay, fresh optimiser state, smaller batches. Per token, it moves the weights more than stage-1 pretraining did at its end.

3 Is a pure pretraining checkpoint susceptible to EM?

EM appears without instruction data. On the stage-1 checkpoint, the bad-medical steering vector gives 28.3% general MR (113/400), against 2.0% for the good-medical vector ($p = 2 \times 10^{-28}$) and 3.8% without fine-tuning (Figure 1). On the final base model it gives 20.8%. Betley et al. [2025, §4.8] showed EM in a base coder model, and Wang et al. [2025] in helpful-only models. The stage-1 checkpoint goes one step further: its training data contain no instruction data, and we can inspect them.

Susceptibility is in place early. We trained the same vector at six checkpoints (Figure 2). The raw rate climbs from 6.3% at 42B tokens to 28.3% at 5.9T, but coherence climbs with it, from 27% to 90%. Among coherent answers the rate is 24–36% at every checkpoint (Fisher p against the

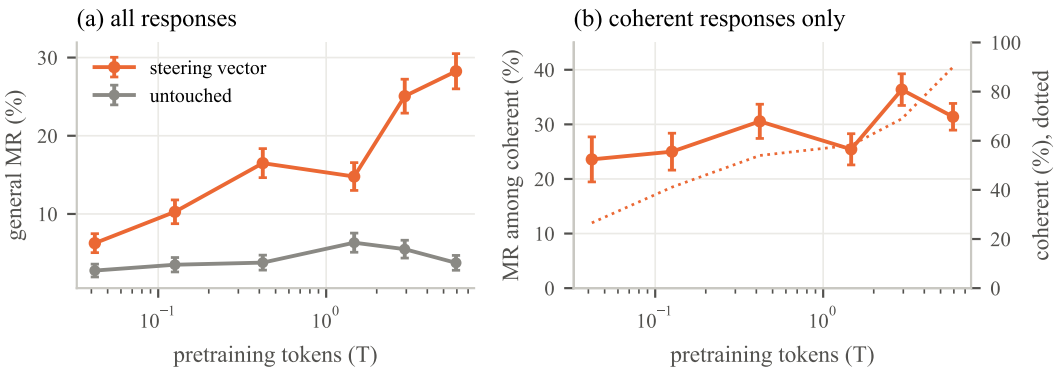


Figure 2: **Once incoherent answers are set aside, susceptibility is flat from 42B tokens onward.** (a) General misalignment rate after training the bad-medical steering vector at six stage-1 checkpoints, from 42B to 5.9T tokens (orange), and without fine-tuning (gray). (b) The same rate among coherent answers; dotted = coherent fraction after steering (right axis). About 400 answers per point, one seed, binomial s.e.

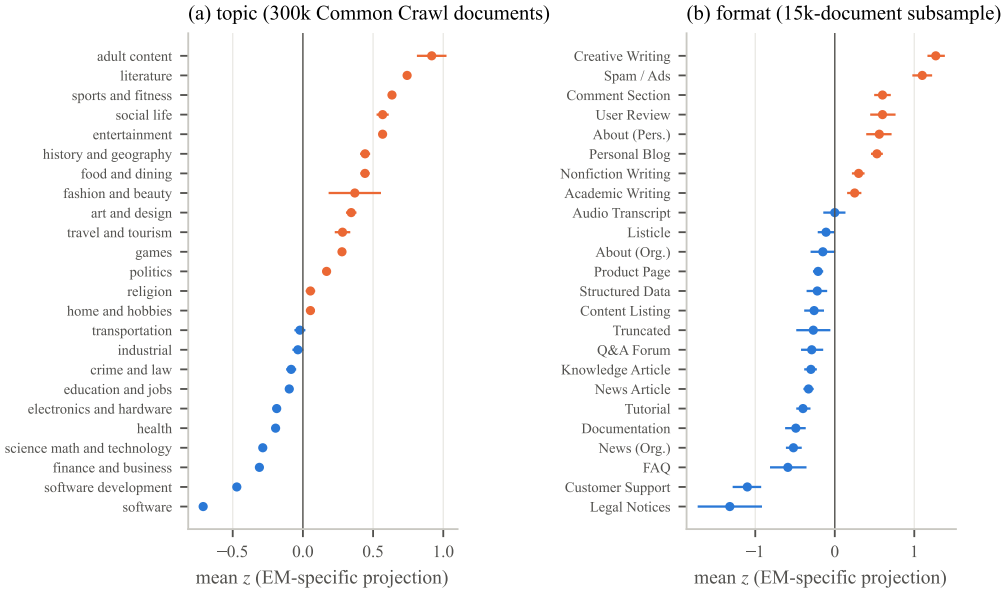


Figure 3: **The EM direction is most active on creative, opinionated and informal text, and least active on technical, legal and instructional text.** Mean EM score (z against 32 random directions; stage-1 model, layer 12) by (a) WebOrganizer topic, over 300k Common Crawl pages, and (b) WebOrganizer format, over a uniform 15k sample of those pages. Bars are 95% CIs over documents.

final checkpoint 0.14–0.85). The model can be pushed into EM as soon as it answers coherently, within the first 0.7% of pretraining. This matches reports that persona vectors form within 0.22% of OLMo-3 pretraining [Moskvoretskii et al., 2026]. Wen et al. [2026] instead report EM rising and falling between OLMo-3 checkpoints. They used insecure-code training and multiple-choice answer probabilities, and six checkpoints cannot rule out such swings between them.

4 What kind of pretraining text does the EM direction respond to?

Topics and formats. Across 300k Common Crawl pages (Figure 3a), adult content ($z = +0.92$), literature ($+0.74$), sports ($+0.63$), social life and entertainment ($+0.57$) score highest. Software (-0.71), software development, finance and science score lowest. Formats separate more sharply (Figure 3b):

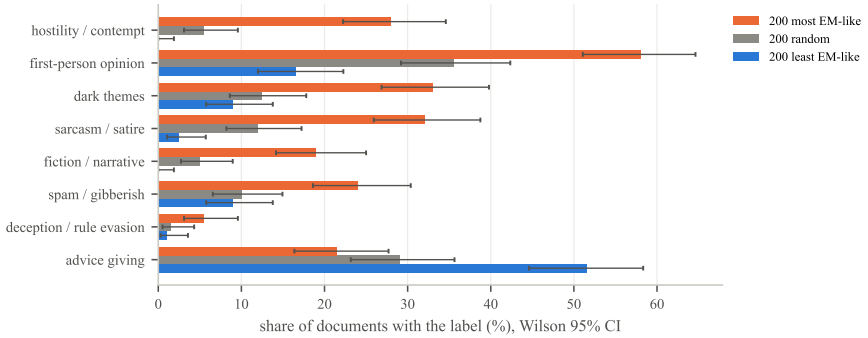


Figure 4: **The most EM-like documents are hostile, sarcastic and opinionated; the least EM-like give advice and instructions.** Share of documents with each label among the 200 most EM-like, 200 random and 200 least EM-like documents of a 45k-document sample, labelled by `gpt-6-luna` without seeing group or score (first 512 tokens). Wilson 95% intervals.

- highest: creative writing ($+1.27$), spam ($+1.10$), comment sections and user reviews ($+0.60$) and personal blogs;
- lowest: legal notices (-1.32), customer support (-1.10), FAQs, news from organisations, documentation and tutorials.

A blind rubric. Topics describe subject matter, but the pattern suggested a voice rather than a subject. We had `gpt-6-luna` label eight properties of the 200 most EM-like, 200 random and 200 least EM-like documents, shuffled and without scores (Figure 4; Bonferroni over the eight labels).

- *Most EM-like vs random:* hostile or contemptuous 28.0% vs 5.5% ($p_{\text{bonf}} = 8.5 \times 10^{-9}$), sarcastic 32% vs 12%, dark themes 33% vs 12.5%, fiction 19% vs 5%, first-person opinion 58% vs 35.5%.
- *Least EM-like vs random:* advice or instructions 51.5% vs 29% ($p_{\text{bonf}} = 5.3 \times 10^{-5}$), with no hostility and little sarcasm.
- *Deception*, the property closest to the training task, is rare in every group (5.5%, 1.5%, 1%).

The labeller marks hostility generously, so absolute rates are upper bounds; the contrast between groups does not depend on that. The pattern matches findings on other models and corpora. The top pretraining documents for GPT-4o's toxic-persona latent are quotes from morally questionable characters [Wang et al., 2025], and EM persona features in four open models attribute to villain and domination narratives in Dolma 3 [Vetter et al., 2026]. The EM direction marks who is speaking and in what tone, more than what the text is about.

5 Does continued pretraining on EM-like text change misalignment?

EM-like text nudges the base model's default behaviour. Table 1 and Figure 5 give the results.

- *EM-like vs no CPT.* After CPT on EM-like documents, general MR rises from 2.6% to 4.3% ($p = 0.076$). Among coherent answers it rises from 5.0% to 9.5% ($p = 0.017$). On the first 400 samples we drew, the rise was larger (3.8% to 8.1%, $p = 0.01$).
- *Ordering.* Misalignment is ordered EM-like (4.3%) > random (3.1%) > least EM-like (2.0%) on the 8 first-plot questions (trend $p = 0.010$; among coherent, $p = 0.024$). On the 48 additional questions the ordering holds (6.5%, 5.9%, 3.5%; trend $p = 0.001$). There, least EM-like text lowers misalignment below no CPT even among coherent answers (12.2% vs 18.1%, $p = 0.024$).
- *Pairwise.* EM-like and random documents do not differ significantly on their own (4.3% vs 3.1%, $p = 0.24$). The effect is an ordering across the three sets, not a large gap between two.

Two confounds come with the content. First, every CPT model loses coherence: 53% of answers are coherent without CPT, against 40–47% after it. The least EM-like set loses the most, so on the first-plot questions its lower raw rate reflects incoherence (5.1% vs 5.0% among coherent answers). Second, EM-like documents are noisier text. Their language-modelling loss is 2.76, against 2.15 for

Table 1: **Continued pretraining on EM-like text raises the base model's misalignment slightly but leaves its susceptibility unchanged.** General misalignment rate (%, misaligned/total) of the stage-1 checkpoint without continued pretraining (CPT) and after 10M tokens of CPT on each document set. "CPT only": the model sampled directly (fresh seed; 100 samples per first-plot question, 25 per additional question). "CPT + EM steering": after training the bad-medical steering vector on the CPT model (50 samples per question). "Coherent answers": % with coherence > 50. Trend p: Cochran–Armitage test for the ordering EM-like > random > least EM-like. One CPT run per set.

	no CPT	EM-like	random	least EM-like	trend p
CPT only, 8 questions	2.6 (21/796)	4.3 (34/795)	3.1 (25/797)	2.0 (16/793)	0.0099
among coherent	5.0 (21/422)	9.5 (34/357)	6.7 (25/372)	5.1 (16/316)	0.024
coherent answers	53.0	44.9	46.7	39.8	
CPT only, 48 questions	7.1 (85/1196)	6.5 (77/1191)	5.9 (70/1190)	3.5 (42/1190)	0.0014
among coherent	18.1 (85/469)	21.8 (77/354)	18.8 (70/373)	12.2 (42/343)	0.001
coherent answers	39.2	29.7	31.3	28.8	
CPT + EM steering, 8 questions	28.2 (113/400)	26.0 (104/400)	25.1 (100/398)	27.0 (108/400)	0.75
coherent answers	90.0	87.2	85.7	88.8	

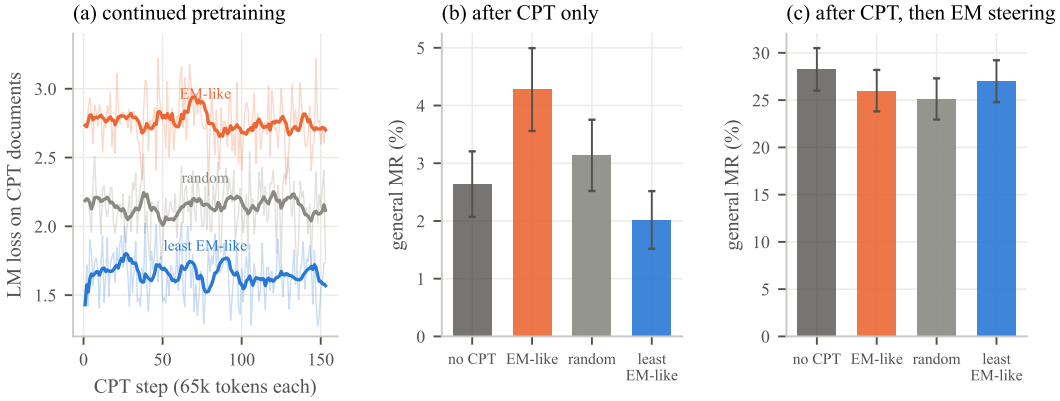


Figure 5: **EM-like text is harder to predict, raises misalignment after CPT alone, and leaves the post-training endpoint unchanged.** (a) Language-modelling loss during 10M tokens of CPT on each document set (faint: per step; solid: 10-step mean). (b) General misalignment rate of each CPT model, 800 answers. (c) After training the bad-medical steering vector on each CPT model, 400 answers. Binomial s.e.

random and 1.65 for least EM-like documents (Figure 5a). Some of the shift may be imitation of an informal, hostile register rather than a change in disposition.

It does not change susceptibility to EM. After the bad-medical steering vector is trained on each CPT model, general MR is 26.0% (EM-like), 25.1% (random) and 27.0% (least EM-like), against 28.2% without CPT (all $p \geq 0.34$; trend $p = 0.75$; Figure 5c). Coherence returns to 86–89% in every model. If EM-like text made the model more susceptible, it might show up as faster convergence rather than a higher endpoint. We therefore evaluated each steering run at six points during training (Figure 6). No CPT model converges faster than the model without CPT. Pooled over the six points, the odds ratios against no CPT are 0.91 (EM-like, $p = 0.33$), 0.70 (random, $p = 0.0008$) and 0.87 (least EM-like, $p = 0.18$) among coherent answers. Ranking documents by a gradient-based score did not change the outcome either. That score is the first-order change in a document's loss when steering along the EM direction, and it is uncorrelated with the activation score (Spearman 0.05). After EM training, 10M tokens of the highest-ranked documents give 22.5%, against 25.1% for random documents ($p = 0.41$).

How this compares with other interventions. Our susceptibility result agrees with every pretraining-stage intervention against EM we know of:

- upsampling alignment discourse to ≈1% of a 550B-token pretraining run leaves models susceptible to the same medical, financial and sports datasets [Tice et al., 2026, App. I];

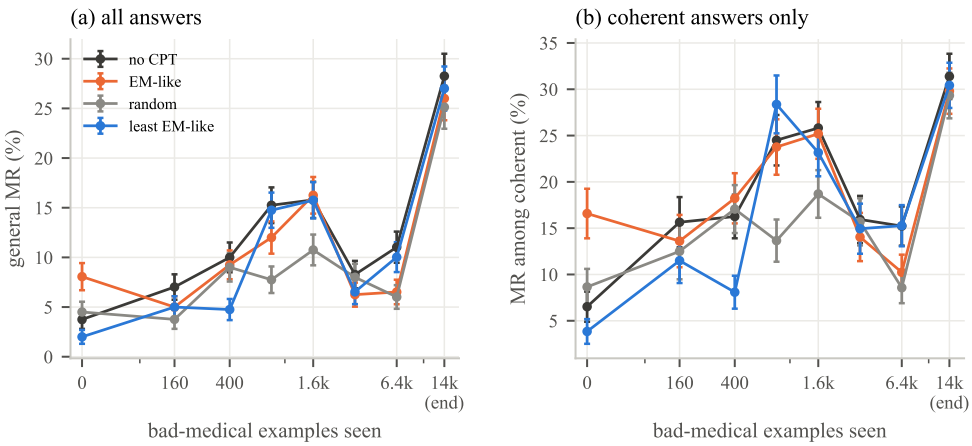


Figure 6: **No continued-pretraining model becomes misaligned faster during EM training.** General misalignment rate during training of the bad-medical steering vector, evaluated before training (0), after 160, 400, 800, 1.6k, 3.2k and 6.4k examples, and at the end (14k), for the stage-1 checkpoint without CPT and after 10M tokens of CPT on each document set. The 0 point uses 50 samples per question like the other points, not the fresh samples of Table 1. (a) All answers; (b) coherent answers only. About 400 answers per point, binomial s.e. All runs share one seed and data order, which gives the curves their common shape.

- synthetic-document midtraining does not prevent EM from reward hacking [Jose and Stastny, 2026];
- 2% conflicting fine-tuning examples override 190M tokens of alignment midtraining [Baines et al., 2026].

Our default-behaviour result complements Vetter et al. [2026]. They fine-tuned instruct models directly on documents attributed to EM persona features and found they did not reliably induce coherent EM. On Gemma 3 27B, their attributed human-written documents gave 4.58% misalignment at 32% incoherence, against 1.67% for random documents, and they conclude that human-written text does not induce coherent EM however it is selected. We continued pretraining a base model on documents selected by its own EM direction. We find a similarly small shift, ordered by the score, that leaves susceptibility untouched. Selected data have moved behaviour in other settings: benign fine-tuning data close to harmful data breaks safety in chat models [He et al., 2024]; a 500M-token final window of curated safety text changes what survives later preference training [Lu et al., 2026]; and ≈250 poisoned documents install a backdoor at any scale [Souly et al., 2025]. Those settings target specific behaviours or curated text, not a disposition built over trillions of tokens.

6 Related work

Emergent misalignment and personas. Betley et al. [2025] discovered EM, including in a base model. Turner et al. [2025] built text datasets and minimal model organisms. Soligo et al. [2026] showed that the general solution is more efficient and stable than a narrow one. The persona account [Wang et al., 2025, Marks et al., 2026, Chen et al., 2025] holds that EM selects a character learned in pretraining. Overlap accounts explain where EM spreads by how similarly the base model represents training and evaluation inputs [Li et al., 2026, Zhang et al., 2026]. We test the persona account's pretraining premise directly, on a model whose corpus and checkpoints are open.

Pretraining data and alignment. Conditioning, curating or filtering pretraining data changes model behaviour [Korbak et al., 2023, Maini et al., 2025, O'Brien et al., 2025]. Discourse about AI in pretraining shifts misalignment in a self-fulfilling way, but does not protect against EM [Tice et al., 2026]. Midtraining interventions are often overridden by later fine-tuning [Baines et al., 2026, Jose and Stastny, 2026]. Persona vectors and refusal-related features form early in pretraining [Moskvoretskii et al., 2026, Malla et al., 2026]. We add a targeted intervention: documents chosen by the model's own EM direction. It moves default behaviour slightly and leaves susceptibility unchanged.

Training-data attribution. Influence functions [Koh and Liang, 2017, Grosse et al., 2023] and cheaper probe- and concept-based variants [Xiao and Aranguri, 2026, Kowal et al., 2026] estimate which training data cause a behaviour. The cheaper variants have been validated on post-training data. SAE-based attribution has been applied to EM features [Vetter et al., 2026]. Our activation score is a first-order relative of these methods, tested here by intervention on pretraining data.

7 Conclusion and limitations

On OLMo-3-7B, EM needs no instruction data, and the model can be pushed into it as soon as it answers coherently, within the first 42B tokens of pretraining. The direction it uses is most active on creative, opinionated and hostile text and least active on instructional and technical text. Continued pretraining on 10M tokens of the most EM-like text makes the base model slightly more misaligned, and misalignment is ordered across EM-like, random and least EM-like text. However, none of these changes how easily the model is pushed into EM. To our question, the answer is partly yes. Pretraining text shapes the model's default behaviour in the direction its content predicts. What makes the model susceptible to EM forms early and is not moved by targeted text at this scale.

Limitations.

- *One run per document set.* Each CPT model is one run with one training seed, so our tests capture sampling noise but not run-to-run variance. The three-way ordering was examined after the pairwise comparisons, and EM-like and random documents do not differ significantly on their own. A larger run with a looser cutoff (100M tokens, the top 11% of a 1.15M-document pool) did not separate EM-like from random documents (5.7% vs 5.5%), so the effect did not grow with more, less selective text.
- *Dose and optimiser.* 10M tokens is 1.7×10^{-6} of stage-1 pretraining, and our optimiser setup differs from OLMo's. A null for susceptibility at this dose does not rule out a data effect at pretraining scale.
- *Coherence and register.* CPT lowers coherence, and EM-like documents are noisier. Part of the rise in misalignment may be register imitation.
- *Judge.* Every rate uses `gpt-6-luna`, an unpinned alias. It agrees with GPT-4o on which answers are misaligned ($\kappa = 0.73$, recall 1.0) but flags about $1.7\times$ as many.
- *Scope.* One model, one EM training task, one layer and one form of EM training (a steering vector).

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A Judge choice and calibration

On JailbreakBench [Chao et al., 2024], 300 responses with human harm labels, agreement with humans was $\kappa = 0.79$ (`gpt-6-luna`), 0.76 (`gpt-4o-2024-08-06`) and 0.81 (`gpt-4.1-mini`). Humans agree with each other at $\kappa = 0.84$. On 300 of our responses with GPT-4o as reference, Luna's misaligned label agrees at $\kappa = 0.73$ with recall 1.0, but it flags 9.0% against 5.4%. The ranking of models is identical. On 70 responses released by Betley et al. [2025], Luna counts 89% as misaligned and coherent and GPT-4o counts 100%.

B Compute and API cost

The numbers below cover the whole project this study was part of, not only the experiments reported here.

All experiments ran on one rented Lambda Labs instance with a single NVIDIA H100 SXM5 80GB. The box was up from 2026-09-29 ≈23:00 UTC to 2026-10-04 ≈17:30 UTC, about 114.5 billed hours. GPU jobs span 105.4 h including 12–15 h of idle gaps between phases, so the GPU was busy for ≈85–95 h. The last 8.4 h were spent uploading artifacts. The judge totals come from token counts. We tokenised all 156,800 judged rows in `results/responses` with `tiktoken` and checked the total against the usage lines `judge.py` printed in the queue logs (384k recorded calls vs. 386k implied by the CSVs). We could not verify `gpt-6-luna`'s price. The range in Table 2 brackets the per-token price implied by two spend notes in our log.

Table 2: **Estimated cost of the project: ≈$510, 95% of it GPU rental.** Low/central/high columns vary the H100 hourly rate ($3.29 vs. $4.29, Lambda list prices for PCIe and SXM) and the unverified `gpt-6-luna` price ($0.05/$0.10/$0.25 per 1M input tokens, $0.40/$0.40/$2 per 1M output tokens). Other OpenAI models are priced at list. Storage (public Hugging Face repo, ≈139 GB; private GitHub) was free. The cost of the Claude sessions used to run the project is not included.

Item	Quantity	Low	Central	High
1× H100 SXM5 80GB (Lambda)	114.5 h billed (≈90 h busy)	$375	$491	$495
Judge, `gpt-6-luna`	386k calls; 138.0M in / 1.55M out tok.	$7.5	$14.4	$39.6
Document labels, `gpt-6-luna`	600 calls; 0.44M in tok.	$0.04	$0.06	$0.19
Judge calibration (4 judges)	4,772 calls	$1.42	$1.45	$1.52
Five-judge comparison, Phase 0	8,000 calls	$0	$3.0	$3.0
Batch API (test only; GPT-4o headline never run)	10 requests	$0	$0	$0
API subtotal		$9	$19	$44
Total		**$385**	**$510**	**$540**

Uncertainty. The low/high range covers the hourly rate and the judge price. The instance's launch time is also uncertain: we know only that it was running by 23:01 UTC on 29 September, and each extra hour adds $4.29. The OpenAI usage dashboard would give the exact API figure.